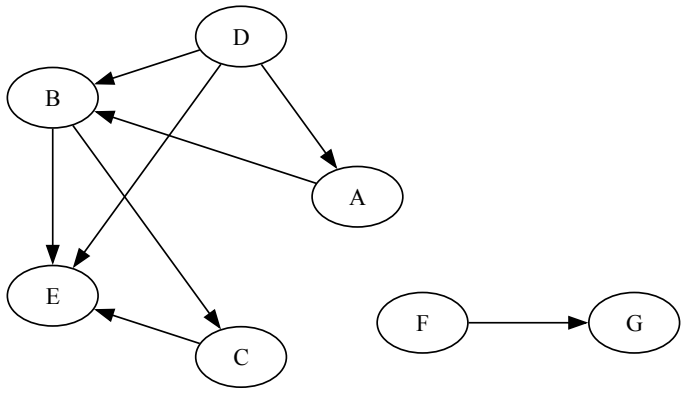
Name _____

Quiz 8

1. (3 points) Select all valid topological orders for the graph below.



   - $ABCDEFG$
   - $GFEDCBA$
   - $DAFBECG$
   - $FDGABEC$
   - FG

2. (3 points) Which type of graph has the larger number of distinct topo orders?

   - A directed graph with n nodes and 0 edges
   - A directed tree with n nodes and $n-1$ edges
   - A directed graph with n nodes and $n \cdot (n+1)/2$ edges (a complete garph) but no cycles

3. (4 points) Recall the minimum coins problem from earlier in the course: *Compute the minimum number of coins needed to make change for n, given that the coins are of denominations 1, 3, 4*. We solved this by defining a recursive definition for the subproblems: Let $MC(i)$ be the minimum number of coins needed to make change for i. Then the recursive definition is:

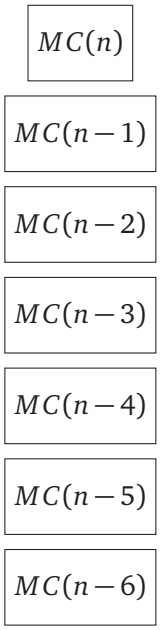
$$MC(i) = \begin{cases} 1 & if\, i = 1,3,4, \\ 2 & if\, i = 2, \\ 1 + \min\{MC(i-1), MC(i-3), MC(i-4)\} & i > 0. \end{cases} \quad (1)$$

   We have previously constructed the *recursion tree* for a call to $MC(n)$; a recursion tree has a node for every *call* and an edge from a call to its child calls.

We will now construct a *dependency graph* for a call to $MC(n)$; here, there is a node for every *subproblem* and an edge from subproblem u to subproblem v if evaluating u depends on the answer to v.

For example, evaluating $MC(n)$ depends on $MC(n-1)$, so there should be a directed edge from $MC(n)$ to $MC(n-1)$ in the dependency graph.

(a) Fill in all of the edges in this example (assume no nodes are base cases):



(b) Now consider memoizing this recursion into an array M. We know we can fill M from left to right, but now let's think about filling it in an order determined by some graph structure in the dependency graph G. Fill in the blank:

We fill M in a _____ of G.

4. There are only four more weeks in the semester and it's time to make sure you have the best chance to get the grades you want! If you anticipate needing or wanting to attend office hours before the semester ends, whether to redo quizzes/homeworks or to get help with course material or exam prep, please write down any additional times you would like to request and I will do my best to add them to the current schedule (Mon 11-12, Fri 9-10).